

Auteur: Rubin Cuypers
Studierichting: Informatica
Oefeningenreeks 1C nr. 46.2

$$z^4 = -1 = \operatorname{cis} \pi \Rightarrow z = \operatorname{cis} \frac{\pi + k2\pi}{4}$$

$$z_0 = \operatorname{cis} \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \quad (k = 0)$$

$$z_1 = \operatorname{cis} \frac{3\pi}{4} = -\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \quad (k = 1)$$

$$z_2 = \operatorname{cis} \frac{5\pi}{4} = -\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \quad (k = 2)$$

$$z_3 = \operatorname{cis} \frac{7\pi}{4} = \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \quad (k = 3)$$

References